

Ordinal Feature Engineering Documentation

Ames Housing Dataset — Structural Variable Mapping Schema

Purpose: This document outlines the explicit mathematical mapping schemas for ordinal categorical variables. Machine learning models (especially linear/distance-based models) struggle with raw string inputs. Converting these qualitative attributes into explicit ordered sequences guarantees that the model natively respects the logical progression of structural quality, wear-and-tear, and feature utility.

When applying standard mapping transformations (such as `pd.Categorical.codes`), the sequential categories are transformed into zero-indexed integers, where the baseline absence of a feature maps to `0`, scaling sequentially up to the highest quality designation.

1. General Quality & Condition Scale (QUALITY_ORDER)

Applied to multiple core attributes including exterior components, basements, heating systems, kitchens, fireplaces, garages, and pool quality metrics.

Abbreviation	Meaning	Numeric Code	Description
None	None	0	Feature does not exist on the property (e.g., No Pool, No Fireplace).
Po	Poor	1	Severe deficiencies; component requires complete replacement.
Fa	Fair	2	Functional but exhibits marked wear, damage, or structural obsolescence.
TA	Typical / Average	3	Normal product for its age; standard utility and condition.
Gd	Good	4	Above-average condition; visually appealing with minor signs of wear.
Ex	Excellent	5	Exceptional quality, luxury material grade, or completely pristine condition.

2. Garage Interior Finish Scale (GARAGE_FINISH_ORDER)

Measures the level of completion and interior infrastructure inside the property's primary vehicle enclosure.

Abbreviation	Meaning	Numeric Code	Description
None	No Garage	0	The property lacks a garage attachment or detached structure.
Unf	Unfinished	1	Bare framing studs, exposed ceiling rafters, and unpainted/un-taped drywall.
RFn	Rough Finished	2	Drywall is mounted and joint-taped, but lacks final texture, paint, or trim.
Fin	Finished	3	Fully sheetrocked, painted, insulated, and equipped with standard lighting/utility.

3. Basement Exposure Scale (BSMT_EXP_ORDER)

Quantifies the entry of daylight and the presence of direct garden/walk-out access for underground levels.

Abbreviation	Meaning	Numeric Code	Description
None	No Basement	0	The structure is built directly on a concrete slab or shallow crawlspace.
No	No Exposure	1	Subterranean walls with either no windows or standard subterranean window-wells.
Mn	Minimal Exposure	2	High-set partial basement windows allowing low levels of natural light.
Av	Average Exposure	3	Standard-sized windows typically found in split-level or daylight basement layouts.
Gd	Good Exposure	4	Full grade daylight windows and/or walk-out grade doors opening to the exterior.

4. Fence Quality & Privacy Scale (FENCE_ORDER)

Evaluates the structural material grade and visual barrier efficacy of the backyard fencing boundary.

Abbreviation	Meaning	Numeric Code	Description
None	No Fence	0	No perimeter boundary fencing exists on the lot.
MnWw	Minimum Wood/Wire	1	Basic wire mesh or sparse wood stakes; functional only for basic boundary definition.
GdWo	Good Wood	2	Traditional solid wood construction or standard picket layout in well-maintained status.
MnPrv	Minimum Privacy	3	Chain-link or semi-open slatted styles that define space but allow high visibility.
GdPrv	Good Privacy	4	High-grade, solid, tall wood, composite, or vinyl panels ensuring complete visual privacy.